Logic answer key (version 1.0.0)

Private grading material. Each entry gives the question ID, its page in the question collection, the correct response, the full reasoning, why each wrong choice fails, accepted alternatives, the grading rule and how the key was verified.

S1-M-Q1 · question page 2

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

The shaded shape is reflected in the dashed mirror line on its right. Which figure shows its mirror image? (Only the shape matters, not where it sits in its box.)

Answer: C. Figure C

Reasoning: A reflection in a vertical line keeps every cell in its row and swaps left and right. Figure C is the only panel whose rows match the shape row by row with left and right swapped.

- ✗ Figure A: a half-turn of the shape, not a reflection
- ✗ Figure B: the reflection in a horizontal line (top and bottom swapped)
- ✗ Figure D: a quarter-turn of the shape

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S1-M-Q2 · question page 3

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

Each figure shows a shape above a dashed water line and, below the line, what is supposed to be its reflection in the water. In which figure can the lower shape NOT be the reflection of the upper one?

Answer: D. Figure D

Reasoning: A reflection in a horizontal line keeps every cell in its column and reverses the order of the rows, measured from the line. In D the lower

shape is the upper shape moved down without flipping, so the row nearest the line is wrong.

- ✗ Figure A: the lower shape is the upper shape flipped top to bottom across the line, so it is a real reflection
- ✗ Figure B: the lower shape is the upper shape flipped top to bottom across the line, so it is a real reflection
- ✗ Figure C: the lower shape is the upper shape flipped top to bottom across the line, so it is a real reflection

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S1-M-Q3 · question page 4

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

Words are written left to right in block capitals and seen in a left-right mirror (a vertical mirror line). In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same in the mirror?

Answer: A. TOOT

Reasoning: A left-right mirror reverses the order of the letters and mirrors each letter. The word looks the same only if it reads the same backward AND every letter is a listed symmetric letter. TOOT is the only choice that does both.

- ✗ NOON: N is not one of the listed letters, so each N appears reversed in the mirror
- ✗ ZOOM: Z is not listed, and the mirror reverses the letter order, giving MOOZ
- ✗ BIB: B is not listed, so each B appears reversed even though BIB reads the same backward

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the

check function accepts only the keyed choice.

S1-M-Q4 · question page 5

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

A word is written in block capitals and seen reflected in calm water below it (a horizontal mirror line). In the block capitals used here, exactly these letters look the same in a top-bottom mirror: B C D E H I K O X. The letter order stays the same in this reflection. Which word looks exactly the same in the water?

Answer: B. CODE

Reasoning: A top-bottom reflection keeps the order of the letters and flips each letter. The word looks the same when every letter is a listed letter; only CODE qualifies.

- ✗ CORE: R is not one of the listed letters, so it appears upside down
- ✗ BEAD: A is not listed, so it appears upside down
- ✗ BOXES: S is not listed, so it appears changed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S1-M-Q5 · question page 6

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

The grid has 4 rows and 8 columns. The dashed mirror line runs between columns 4 and 5. The shaded cells are in row 1 column 2, row 2 columns 2, 3 and 4, and row 3 column 4. Mark the cells of their mirror image. (Cells are named row,column; row 1 is the top row and column 1 the left column.)

Answer: 1,7, 2,5, 2,6, 2,7, 3,5

Reasoning: Each cell keeps its row, and a cell in column c lands in column 9 - c (the same distance on the other side of the line between columns 4 and 5): (1,7), (2,7), (2,6), (2,5), (3,5).

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: the key is computed as column 9 - c for every shaded cell.

S1-B1-Q1 · question page 7

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A grid has 10 columns. A vertical mirror line lies between columns 5 and 6. A shaded cell is in column 2. In which column is its mirror image?

Answer: 9

Reasoning: Column 2 is 4 columns left of the line (columns 2, 3, 4, 5), so its image is 4 columns right of it: column 9. Check: 2 + 9 = 11 = 5 + 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-B1-Q2 · question page 8

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A grid has 8 rows. A horizontal mirror line lies between rows 4 and 5. A shaded cell is in row 1. In which row is its mirror image?

Answer: 8

Reasoning: Row 1 is 4 rows above the line, so the image is 4 rows below it: row 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-B1-Q3 · question page 9

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A reflection in a vertical mirror line keeps every cell in its own row.

Answer: True

Reasoning: Only the column changes; the row stays the same.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S1-B1-Q4 · question page 10

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A cell is 3 columns to the left of a vertical mirror line. Where is its mirror image?

Answer: D. 3 columns to the right of the line

Reasoning: A reflection keeps the distance to the line and changes the side.

- ✗ 3 columns to the left of the line: that is the original cell
- ✗ 6 columns to the right of the line: the distance from the line is kept, not doubled
- ✗ on the mirror line itself: only points on the line stay on it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S1-B1-Q5 · question page 11

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A row of 6 cells has a mirror line between cells 3 and 4. Cell 1 is shaded. Mark the cell of its mirror image. (The grid is 1 row by 6 columns; cells are named row,column.)

Answer: 1,6

Reasoning: Cell 1 is 3 cells left of the line, so its image is 3 cells right: cell 6.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: image column = 7 - 1.

S1-F1-Q1 · question page 12

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A clock face with no numbers shows 2:20. You see it in a left-right mirror. What time does the mirrored clock appear to show?

Answer: C. 9:40

Reasoning: A left-right mirror sends a time t to 12:00 - t. 12:00 - 2:20 = 9:40. Check: the minute hand at 4 (20 minutes) appears at 8 (40 minutes), and the hour hand just past 2 appears just before 10.

- ✗ 10:20: only the hour hand was mirrored; the minute hand was left at 20 minutes
- ✗ 2:40: only the minute hand was mirrored
- ✗ 3:40: the hands were swapped instead of mirrored

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S1-F1-Q2 · question page 13

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

The shape is reflected in a vertical mirror line and then the result is reflected in a horizontal mirror line. Which figure shows the final result? (Only the shape matters, not where it sits.)

Answer: B. Figure B

Reasoning: A vertical reflection then a horizontal one sends (x, y) to (-x, -y): a half-turn. Figure B is the shape turned halfway around.

- ✗ Figure A: only the first (vertical) reflection
- ✗ Figure C: the starting shape: the two reflections do not cancel
- ✗ Figure D: only the second (horizontal) reflection

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S1-F1-Q3 · question page 14

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A word is written DOWN the page, one block capital under another, and seen in a left-right mirror. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same?

Answer: A. MATH

Reasoning: Written downward, a left-right mirror keeps the order (top to bottom) and mirrors each letter, so the word looks the same when every letter is listed. Only MATH qualifies.

- ✗ MAZE: Z and E are not listed, so they appear reversed
- ✗ BATH: B is not listed
- ✗ MASK: S and K are not listed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S1-F1-Q4 · question page 15

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A 6 by 6 grid has two dashed mirror lines: a vertical one between columns 3 and 4 and a horizontal one between rows 3 and 4. The figure must look the same in BOTH mirrors. Four cells are already shaded in the top-left quarter: (1,1), (1,2), (2,2) and (3,3). Mark every shaded cell of the completed figure, including the four given cells.

Answer: 1,1, 1,2, 1,5, 1,6, 2,2, 2,5, 3,3, 3,4, 4,3, 4,4, 5,2, 5,5, 6,1, 6,2, 6,5, 6,6

Reasoning: Each given cell (r, c) also needs its images (r, 7 - c), (7 - r, c) and (7 - r, 7 - c). That gives 4 x 4 = 16 cells, one copy of the pattern in

each quarter.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: the key is the union of the four images of each given cell.

S1-F1-Q5 · question page 16

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A figure looks the same after a reflection in a vertical line through its center and after a reflection in a horizontal line through its center. Then it also looks the same after a half-turn about its center.

Answer: True

Reasoning: Reflecting in the vertical line and then in the horizontal line sends (x, y) to (-x, y) and then to (-x, -y), which is a half-turn. Each reflection leaves the figure unchanged, so the half-turn does too.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S1-F2-Q1 · question page 17

LOGIC.SPATIAL.01 · Construct and count symmetric figures

The figure is on a 4 by 4 grid with a dashed vertical line between columns 2 and 3. Shaded cells can be added but not removed. What is the least number of cells you must add so that the figure looks the same in a mirror along the dashed line?

Answer: 5

Reasoning: For symmetry every shaded cell needs its image across the line (column c pairs with column 5 - c). Row 1: columns 1 and 2 need columns 4 and 3. Row 2: column 1 needs column 4; columns 2 and 3 are already each other's images. Row 3: column 1 needs column 4. Row 4: column 2 needs column 3. New cells: (1,3), (1,4), (2,4), (3,4), (4,3), so 5. Adding fewer leaves some shaded cell without its image.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-F2-Q2 · question page 18

LOGIC.SPATIAL.01 · Construct and count symmetric figures

Strings of 3 letters are made from the letters A, H, M, O and T (letters may repeat). Each is written left to right in block capitals. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. How many of these strings look exactly the same in a left-right mirror?

Answer: 25

Reasoning: All five letters are mirror-symmetric, so a string looks the same exactly when it reads the same backward: the first and last letters match. Choose the first letter (5 ways) and the middle letter (5 ways): 25.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-F2-Q3 · question page 19

LOGIC.SPATIAL.01 · Construct and count symmetric figures

Two vertical mirror lines are 3 cells apart: line 1 on the left, line 2 on the right. A shape is reflected in line 1 and the image is then reflected in line 2. Where does the final image end up?

Answer: D. the original shape moved 6 cells to the right, not flipped

Reasoning: Put line 1 at x = a and line 2 at x = b = a + 3. The first reflection sends x to 2a - x; the second sends that to 2b - (2a - x) = x + 2(b - a) = x + 6. Every point moves 6 to the right and nothing is flipped.

✗ the original shape moved 3 cells to the right, not flipped: two reflections move the shape twice the distance between the lines

✗ a mirror image of the shape, 6 cells to the right: two reflections undo each other's flip

✗ back exactly where the shape started: that only happens when both lines are the same line

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S1-F2-Q4 · question page 20

LOGIC.SPATIAL.01 · Construct and count symmetric figures

A row of 12 cells is numbered 1 to 12. A mirror line lies between cells 6 and 7. Give two cell numbers a and b, with a less than b, that are mirror images of each other.

Answer: any values that satisfy: a + b == 13 and a < b and a >= 1 and b <= 12 and isInteger(a) and isInteger(b) (example: a = 4, b = 9)

Reasoning: Cells that mirror each other are the same distance from the line, so their numbers add to 6 + 7 = 13: (1, 12), (2, 11), (3, 10), (4, 9), (5, 8) or (6, 7). Any one pair is correct.

Tempting answers the rule rejects: a = 4, b = 10; a = 6, b = 6; a = 9, b = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 6 valid answers.

S1-F2-Q5 · question page 21

LOGIC.SPATIAL.01 · Construct and count symmetric figures

Count only mirror lines that run vertically, horizontally or diagonally through the center of the shape (the middle of the smallest rectangle around it). Which figure has exactly ONE mirror line?

Answer: A. Figure A

Reasoning: Testing each kind of line: A (a T shape) looks the same only in the vertical line. B looks the same after a half-turn but in no mirror; C and D each have four mirror lines.

- ✗ Figure B: no mirror line: it only looks the same after a half-turn
- ✗ Figure C: four mirror lines (vertical, horizontal and both diagonals)
- ✗ Figure D: four mirror lines (vertical, horizontal and both diagonals)

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.